

Show your work. All problems are worth 4 point unless stated otherwise.

1. Write the expression, $x + x^2 + x^3 + x^4 + \cdots + x^{20}$ in sigma notation.

$$\sum_{n=1}^{20} x^n$$

2. Write out the following sum $\sum_{n=0}^3 n^2$.

$$\sum_{n=0}^3 n^2 = 0 + 1^2 + 2^2 + 3^2 = 1 + 4 + 9 = 14$$

Evaluate the geometric series or state that it diverges.

3. $\sum_{k=0}^{\infty} \frac{1}{4^k}$

$$\sum_{k=0}^{\infty} \frac{1}{4^k} = \sum_{k=0}^{\infty} 1 \left(\frac{1}{4}\right)^k = \frac{1}{1 - 1/4} = \frac{4}{3} \text{ since } r = 1/4 < 1.$$

4. $\sum_{m=0}^{\infty} \frac{(-3)^{2m}}{5^m}$

$$\sum_{m=0}^{\infty} \frac{(-3)^{2m}}{5^m} = \sum_{m=0}^{\infty} \left(\frac{9}{5}\right)^m = \infty \text{ since } r = 9/5 > 1$$

Using the Squeeze Theorem, find the limit of the following sequence or determine that the limit does not exist.

5. $\lim_{n \rightarrow \infty} \left\{ \frac{(-1)^n}{n} \right\}$

$$\frac{-1^n}{n} \leq \frac{(-1)^n}{n} \leq \frac{1^n}{n} \text{ and } \lim_{n \rightarrow \infty} \left\{ \frac{-1^n}{n} \right\} = \lim_{n \rightarrow \infty} \left\{ \frac{1^n}{n} \right\} = 0 \text{ so by the Squeeze Theorem, } \lim_{n \rightarrow \infty} \left\{ \frac{(-1)^n}{n} \right\} = 0.$$

Use the Divergence Test to determine whether the following series diverge, or state that the test is inconclusive.

6. $\sum_{k=0}^{\infty} \frac{1}{k+1}$

$$\lim_{n \rightarrow \infty} \frac{1}{n+1} = 0, \text{ so then the test is inconclusive.}$$

7. Let $f(x) = 3e^x$. (Each part is worth 4 points.)

a. Find the Taylor polynomial of order $n = 2$ for $f(x)$ centered at 0.

$f'(x) = 3e^x$ and $f''(x) = 3e^x$. So then

$$\begin{aligned} p_2(x) &= f(0) + f'(0)(x) + \frac{f''(0)}{2}(x)^2 \\ &= 3 + 3x + \frac{3}{2}x^2 \end{aligned}$$

b. Use the Taylor polynomial from above to approximate $3e^{0.1}$, and compute the absolute error in the approximation assuming the exact value is given by a calculator.

$$p_2(.1) = 3 + 3(.1) + \frac{3}{2}(.1)^2 = 3.315 \text{ and } 3e^{0.1} \approx 3.31551. \text{ So } |3e^{0.1} - 3.315| \approx .000513$$

Give a set of parametric equations that describes the curve.

8. Give two different sets of parametric equations that generate a circle centered at $(0, 0)$ with radius 5.

a. [2 pts] Set one:

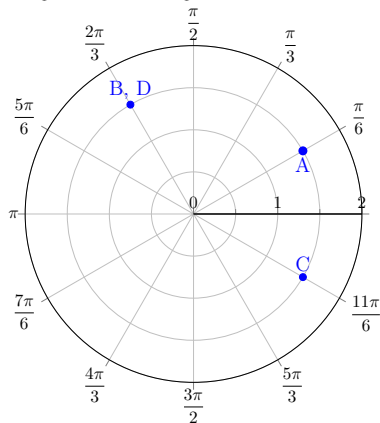
There are many answers: $x = 5 \cos t$ and $y = 5 \sin t$

b. [2 pts] Set two:

There are many answers: $x = 5 \cos(-t)$ and $y = 5 \sin(-t)$

Locate the following polar coordinates on a graph.

9. A: $(1.5, \frac{\pi}{6})$, B: $(1.5, \frac{\pi}{6} + \frac{\pi}{2})$, C: $(1.5, -\frac{\pi}{6})$, D: $(-1.5, -\frac{\pi}{6})$



Express the following **polar** coordinates in **Cartesian** coordinates.

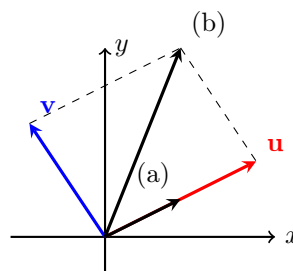
10. $\left(2, \frac{\pi}{3}\right)$

$$x = 2 \cos \pi/3 = 1 \text{ and } y = 2 \sin \pi/3 = \sqrt{3}$$

$$\text{coordinates: } (1, \sqrt{3})$$

Use the graph to draw the following vectors (provide labels). Make certain to indicate direction.

11. a. [2 pts] $\frac{1}{2}\mathbf{v}$
 b. [2 pts] $\mathbf{u} + \mathbf{v}$



Use the vectors $\mathbf{u} = \langle -1, 1, 1 \rangle$ and $\mathbf{v} = \langle 2, -4, 0 \rangle$ to complete the following problems.

12. $\mathbf{u} \cdot \mathbf{v}$

$$\langle -1, 1, 1 \rangle \cdot \langle 2, -4, 0 \rangle = -1(2) + 1(-4) + 1(0) = -2 - 4 = -6$$

13. Find the angle between $\mathbf{u}$ and $\mathbf{v}$. ($|\mathbf{u}| = \sqrt{3}$ and $|\mathbf{v}| = 2\sqrt{5}$.)

$$\mathbf{u} \cdot \mathbf{v} = -6 = \sqrt{3}2\sqrt{5} \cos \theta \Rightarrow \theta = \cos^{-1} \left(-\frac{3}{\sqrt{15}} \right) \approx 2.457$$

14. Find a vector perpendicular to $\mathbf{u}$ and $\mathbf{v}$.

$$(\text{answers may vary}) \mathbf{u} \times \mathbf{v} = \det \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -1 & 1 & 1 \\ 2 & -4 & 0 \end{vmatrix} = 4\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$$

$$\mathbf{n} = \langle 4, 2, 2 \rangle$$

15. A woman paddles a canoe at 5 mi/hr at 30 degrees ($\frac{1}{6}\pi$ radians) north of east. Find a vector describing the vertical and horizontal components of the canoe's velocity.
 $\langle 5 \cos(\pi/6), 5 \sin(\pi/6) \rangle = \langle 5\sqrt{3}/2, 5/2 \rangle$

Find the indicated partial derivative of the following functions.

16. Find f_x and f_y for $f(x, y) = 5x \cdot \sin 2xy$.
 $f_x = 5 \sin 2xy + 10xy \cdot \cos 2xy$
 $f_y = 10x^2 \cos 2xy$

17. Find the four second partial derivatives of $f(x, y) = 2x^5y^2 + 2x^2$.
 $f_x = 10x^4y^2 + 4x$ and $f_y = 4x^5y$, so
 $f_{xx} = 40x^3y^2 + 4$, $f_{yy} = 8x^5$, and $f_{xy} = f_{yx} = 20x^4y$

Use the Chain Rule to find the following derivative. When feasible express your answer in terms of the independent variable.

18. dW/dt where $W = x^2 + y^2$, $x = \cos 2t$, and $y = \sin 2t$.

$$\frac{dW}{dt} = \frac{dw}{dx} \frac{dx}{dt} + \frac{dw}{dy} \frac{dy}{dt} = (2x)(-2 \sin 2t) + (2y)(2 \cos 2t) = (2 \cos 2t)(-2 \sin 2t) + (2 \sin 2t)(2 \cos 2t) = 0$$

Compute the gradient of the following function and evaluate it at the given point P .

19. $f(x, y) = e^{2xy}$; $P(2, 3)$
 $f_x = 2ye^{2xy}$ and $f_y = 2xe^{2xy}$, so $\nabla f(2, 3) = \langle 2ye^{2xy}, 2xe^{2xy} \rangle|_{(2,3)} = \langle 4e^{12}, 6e^{12} \rangle$

Let $f(x, y) = \ln(xy)$ and $P(2, 3)$ be a point on f .

20. Find a vector that points in a direction of steepest ascent in f at P .

Any vector in the direction of: $\nabla f(2, 3) = \left\langle \frac{1}{x}, \frac{1}{y} \right\rangle|_{(2,3)} = \left\langle \frac{1}{2}, \frac{1}{3} \right\rangle$

21. Find a vector that points in a direction of no change in f at P .

Any vector parallel to: $\left\langle -\frac{1}{3}, \frac{1}{2} \right\rangle$

Use the Second Derivative Test to determine (if possible) whether each critical point of the given function is a local maximum, local minimum, saddle point, or inconclusive.

22. [8 pts] $g(x, y) = x^4 + y^4 - 16xy$ with critical points $(0, 0)$, $(2, 2)$, and $(-2, -2)$.

$(0, 0)$: $D_g = g_{xx}g_{yy} - g_{xy}^2 = (12x^2)(12y^2) - (-16)^2 = 144x^2y^2$

$D_g(0, 0) = -16^2 < 0$ so f has a saddle point at $(0, 0)$

$(2, 2)$: $D_g(2, 2) = 12^2 16 - 16^2 > 0$ and $g_{xx}(2, 2) = 12(4) > 0$ so $g(2, 2)$ is a local minimum.

$(-2, -2)$: $D_g(-2, -2) = 12^2 16 - 16^2 > 0$ and $g_{xx}(-2, -2) = 12(4) > 0$ so $g(-2, -2)$ is a local minimum.

BONUS

Use Lagrange multipliers to find the maximum and minimum values of f (when they exist) subject to the constraint.

23. $f(x, y) = e^{2xy}$ subject to $x^3 + y^3 = 16$

24.